ACCESS · INDIGENOUS SKILLS 4 SUCCESS

# Teaching and answer guidance · Materials, plants and quality evidence

Volume 3 · C1, C2, C3, C4, D1, D2, D3, E1, E2, E3, E4

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## In this volume


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# Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for materials, plants and quality evidence. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

## Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

## Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

## Keep the term with its meaning

**AGGREGATE**
Mineral particles forming the solid skeleton of the mixture or granular layer.

**BINDER**
The asphalt material that coats and binds the aggregate in the specified mixture.

**COMPACTED THICKNESS**

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

**DENSITY REFERENCE**
The specified comparison value used as the denominator of a percent-density calculation.

**GRADATION**
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

**HEAD OF MATERIAL**
The quantity/distribution of mix immediately in front of the screed.

**JOB MIX FORMULA (JMF)**
The accepted mixture proportions and properties under the applicable quality/contract process.

**LOCKOUT**
Isolation, securing and verification of hazardous energy for work requiring protection.

**LOT / SUB-LOT**
Defined portions of work used for sampling, testing and acceptance under the contract.

**OMM**
Operation and maintenance manual matched to the actual machine, serial range and configuration.

**RAP**
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

**SDS**
Safety data sheet for the actual product; read with its label and workplace procedure.

**SCREED**
The paver assembly that forms the mat; its response depends on setup, material and movement.

**STATION / OFFSET**
Location along an alignment / distance and direction from a defined reference.

**TCP**
Traffic control person with the required training and assigned role.

**TRACEABILITY**
A record connecting material, location, time, sample, result and authorized decision.

**SVG**
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

**DOCX / XLSX**
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

**PDF / HTML**
Page-based reading copy / browser document; inspect completeness, access and linked resources.

### MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

# Crushing process and paving aggregate production

## Worked example · Find the process change

Fictional circuit: a screen sends its oversize back to a cone crusher. A damaged screen allows larger particles into the finished-product pile. The delivery ticket still carries the correct product name.

The name on the ticket has not verified the physical product. Identify affected production and stockpile records, isolate the questioned product through the quality process, inspect the screen safely and obtain the specified testing. Do not blend the questioned pile into accepted material to hide the difference. The process diagram points to where the change occurred and which records may be affected.

### Trace a product through feeder, jaw/cone/impact stage, screen, conveyor, stockpile and transport

Trace a product through feeder, jaw/cone/impact stage, screen, conveyor, stockpile and transport. Explain which stages change material and which preserve or separate it.

**Answer guidance.**

Crushers reduce/break material; screens separate size fractions; washing removes specified fines; conveyors/transport move it; stockpile handling preserves identity and limits segregation. Feed controls input.

**Observe:** Reading, Communication through the learner's actual contribution, calculation, explanation or saved work.

### A fictional delivery uses 12 truckloads of 18 t

A fictional delivery uses 12 truckloads of 18 t. Two loads are held after the screen fault is discovered. Calculate the recorded total and the quantity held, without calling the remainder accepted.

**Answer guidance.**

12 × 18 = 216 t recorded; 2 × 18 = 36 t held. The other 180 t still needs the applicable quality decision; arithmetic does not establish its conformity.

**Observe:** Numeracy, Problem solving, Writing through the learner's actual contribution, calculation, explanation or saved work.

### Create a digital material-route record for the worked case

Create a digital material-route record for the worked case. Ask a partner to trace affected loads back to the pile and production interval; improve any missing identifiers.

**Answer guidance.**

Record contains source/product/pile, production time, tickets and hold/review status. A successful peer trace uses identifiers rather than memory. Retain the revision and individual explanation.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Observe the actual process from authorized locations under site induction. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess identification of crushers, screens, feeders, conveyors, stockpiles and transport interfaces, plus the learner's explanation of product identity. | 0 / 1 / 2 | Evidence/help: _____ |
| Sample handling and machine service require their own safe procedures and authorized personnel. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Types and specifications of aggregates

## Worked example · Read a sieve result

Fictional dry sample mass is 1,000 g. The measured retained masses on three successive sieves are 150 g, 350 g and 300 g; the pan holds 200 g. Use these values only to practise the arithmetic.

The masses total 1,000 g. Cumulative retained percentages are 15%, 50% and 80%; cumulative passing values are 85%, 50% and 20%. The pan fraction is 20%. Whether the product conforms cannot be decided without sieve sizes, specified method, product identity and acceptance limits.

### Make a product selection table for bedding, drainage, asphalt mixture and concrete sand

Make a product selection table for bedding, drainage, asphalt mixture and concrete sand. Name the information needed before accepting a substitution.

**Answer guidance.**

Use the intended product specification, grading/quality tests, source and application. Product nicknames or appearance alone are insufficient.

**Observe:** Reading, Writing, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

### Recalculate the example if 20 g is missing from the total

Recalculate the example if 20 g is missing from the total. Then calculate PI for fictional LL = 32 and PL = 21. Explain the limits of both results.

**Answer guidance.**

A 980 g recovery triggers the specified mass-reconciliation check; do not silently normalize an unexplained loss. PI = 11 for the tested fine fraction; it does not determine whole-aggregate suitability alone.

**Observe:** Numeracy, Communication through the learner’s actual contribution, calculation, explanation or saved work.

### Plot the supplied cumulative passing values with clearly labelled sieve positions

Plot the supplied cumulative passing values with clearly labelled sieve positions. Save the chart, let a partner find the missing real sieve sizes, and revise its title to avoid implying acceptance.

**Answer guidance.**

Chart preserves 85/50/20% and explicitly identifies a fictional arithmetic exercise. Missing sieve sizes prevent a real gradation comparison; actual digital plotting/save is observed.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use labelled samples and the approved test method under supervision; do not identify a product by colour alone. | 0 / 1 / 2 | Evidence/help: _____ |
| Observe correct terminology, sample/product traceability and interpretation of a sieve table. | 0 / 1 / 2 | Evidence/help: _____ |
| Laboratory competence and approval of a source/product require the applicable qualified testing and decision process. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Asphalt plant operation

## Worked example · One product, two process records

Fictional batch card: 950 kg aggregate plus 50 kg binder, with no other mass included. Fictional continuous record: 19,000 kg aggregate plus 1,000 kg binder over an interval. Compare binder mass percentage on a total-mixture basis.

Batch total = 1,000 kg; binder = 5%. Continuous total = 20,000 kg; binder = 5%. The same arithmetic ratio does not establish matching gradation, moisture correction, binder grade, temperature or conformity. The records describe different production organizations; both require the approved formula and checks.

### Explain and apply the idea

Trace both plant diagrams and explain hot elevator, screen deck, hot bin, weigh bin, pug mill, slat conveyor, silo and loadout.

**Answer guidance.**

Correctly distinguish the conventional batch path from continuous proportioning. The explanation includes material identity and rejects a single universal drum layout.

**Observe:** Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

### Recalculate the batch if binder is 52 kg while aggregate remains 950 kg

Recalculate the batch if binder is 52 kg while aggregate remains 950 kg. State the denominator and why the result alone cannot approve the product.

**Answer guidance.**

52/(950+52) × 100 ≈ 5.19% by total mixture mass. Approval also requires the actual accepted formula, method and other properties.

**Observe:** Numeracy, Writing, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

### Create a saved side-by-side production record

Create a saved side-by-side production record. A partner introduces a bin-label mismatch; identify affected records and revise the release status.

**Answer guidance.**

The mismatch triggers product/process review; do not correct only the label and assume the material is unchanged. Preserve batch/interval and load traceability, with individual digital and collaboration evidence.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use the actual plant process drawing and authorized observation area. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess component/function identification, input/output traceability and recognition of a changed production condition. | 0 / 1 / 2 | Evidence/help: _____ |
| Plant operation, scale calibration and maintenance are separate authorized practical tasks. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Asphalt products and uses

## Worked example · Resolve the document conflict

Fictional register lists drawing P-04 revision C and specification SP-07 revision 2. A delivery note refers to an older project's MMCD detail. The crew is asked to change the surface material at the entrance.

The delivery note does not establish the governing design. Identify the current project documents, describe the requested change and obtain the authorized design/contract response. Keep the reason and affected drawing/specification revisions together. Do not treat an acronym or a supplier recommendation as approval.

### Explain and apply the idea

Create a five-column authority map: owner, designer, contractor QC, owner QA/acceptance and regulator. Place MMCD/ministry/project documents in their actual roles.

**Answer guidance.**

Roles may be held by different organizations; identify project-specific authority. Standard publishers provide documents, while adoption and decision authority come from the project/legal context.

**Observe:** Reading, Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

### Calculate and check the result

Write a clarification request for the worked case that names the conflict, location, current versions and decision needed.

**Answer guidance.**

Request states P-04 C/SP-07 2, the old-project reference and proposed entrance substitution; seeks an authorized response before changing work.

**Observe:** Writing, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Communicate and verify the work

Build a digital revision register and peer-test whether another learner can find the current entrance requirements. Add a new response and show the superseded record without deleting its history.

**Answer guidance.**

Current/superseded status, issuer, date, adoption and response are visible. Each learner demonstrates save/reopen and a justified improvement to findability.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use a permitted project document set or the complete fictional case. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess correct document selection, interpretation and referral of an unresolved design conflict. | 0 / 1 / 2 | Evidence/help: _____ |
| No classroom interpretation changes a real contract or design. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Environmentally friendly properties

## Worked example · Recycled proportion is not a full impact score

Fictional material record: a 240 t proposed mix contains 48 t of approved RAP. A sales note calls it “20% recycled, therefore 20% lower whole-life emissions.” No emissions or service-life data is supplied.

48/240 × 100 = 20% by mass in this record. That establishes the stated proportion only. Whole-life emissions need comparable quantities, processing/production, transport, construction, maintenance and end-of-life evidence. The proposed reduction is not demonstrated by the recycled percentage.

### Explain and apply the idea

Rewrite the sales note into one supported statement and three questions needed for a whole-life comparison.

**Answer guidance.**

Supported: proposed mixture contains 20% RAP by the stated total mass. Ask for common function/service period, production/haul data and maintenance/end-of-life assumptions, among other relevant needs.

**Observe:** Reading, Writing, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

### Calculate the RAP mass for a second fictional 300 t mix at 15%

Calculate the RAP mass for a second fictional 300 t mix at 15%. Explain why comparing 48 t with the second mass does not rank environmental performance.

**Answer guidance.**

300 × .15 = 45 t. Different total quantities and proportions do not supply equivalent-service impacts; no ranking follows.

**Observe:** Numeracy, Communication through the learner’s actual contribution, calculation, explanation or saved work.

### Create a visual comparison brief with explicit boundaries

Create a visual comparison brief with explicit boundaries. Test it with a partner who asks about runoff or community access, then revise and save it.

**Answer guidance.**

The brief adds relevant non-carbon concerns and information needs without inventing local impacts or permissions. Individual revision and actual digital actions are recorded.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Assess evidence interpretation using documented or fictional product records. | 0 / 1 / 2 | Evidence/help: _____ |
| Check that the learner distinguishes recovery potential, actual reuse and verified environmental outcomes. | 0 / 1 / 2 | Evidence/help: _____ |
| Environmental approvals and material-use decisions remain with the appropriate qualified/authorized processes. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Use of storage areas

## Worked example · Containment is a working system

Fictional inspection record: the containment area is intact, but its drain is open and collected rainwater has not been assessed for contamination. A delivery is due in 30 minutes.

The record does not establish effective containment for the transfer. Have the responsible person implement the storage/transfer procedure, address the open drain and assess the collected liquid before release or transfer. Delivery timing does not change the product or environmental requirements. Record the condition and resolution rather than ticking “tank intact” as the whole inspection.

### Draw the difference between separate containment and an integral self-contained arrangement

Draw the difference between separate containment and an integral self-contained arrangement. Identify five checks both still require.

**Answer guidance.**

Product identity/compatibility, condition, usable capacity, transfer/overfill controls, permitted location and emergency/waste arrangements are relevant. The teaching Type 1/2 labels are not legal certification.

**Observe:** Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

### Calculate and check the result

Fictional cost exercise: virgin material supply costs $8,000; recovered feed costs $5,000 plus $2,400 processing and $900 extra transport. Compare only these stated costs.

**Answer guidance.**

Recovered-route subtotal = $8,300, $300 higher. This bounded estimate does not include every production/performance/life-cycle factor and does not authorize the material.

**Observe:** Numeracy, Writing, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

### Create a digital storage inspection record and peer-test it with the open-drain case

Create a digital storage inspection record and peer-test it with the open-drain case. Revise it to distinguish observation, action, responsible person and verified closure.

**Answer guidance.**

A useful record captures actual condition and evidence rather than a vague safe/unsafe tick. Each learner saves/reopens the revised version and explains the change.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use the actual installation, SDS, transfer plan and permitted observation area. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess product identification, containment/compatibility questions and reporting of a defect. | 0 / 1 / 2 | Evidence/help: _____ |
| Live fuel/binder transfer and burner changes require separate authorized instruction and controls. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Asphalt plant and placement emissions

## Worked example · Two measurements, different questions

Fictional records show a compliant stack sample from last week and a worker complaint about fumes at today's loadout. The supervisor proposes closing the complaint using the stack result.

The records differ in location, time and purpose. The stack result does not resolve today's worker exposure. Apply the workplace reporting/assessment process, control the affected task as required and retain both records with their identities. Investigate the actual loadout conditions and any process change without inventing a concentration.

### Trace dirty gas, filtered gas and collected dust through the baghouse drawing

Trace dirty gas, filtered gas and collected dust through the baghouse drawing. Explain one pollutant-control claim the drawing cannot support.

**Answer guidance.**

Correctly separate gas and solids paths. Fabric dust collection alone does not prove removal of all gases or compliance with every permit/exposure requirement.

**Observe:** Reading, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside

Create a jurisdiction-and-record table for a fictional plant inside Metro Vancouver and one outside. Keep environmental and worker-exposure records in separate columns.

**Answer guidance.**

Inside: current regional air program/actual authorization; outside: applicability of provincial Asphalt Plant Regulation plus other relevant requirements. Workplace exposure control remains a separate question in either case.

**Observe:** Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

### Save a plume/complaint observation form

Save a plume/complaint observation form. A partner adds a change of fuel and a different time interval; revise the questions and explain the missing evidence.

**Answer guidance.**

Record distinguishes observation from measured result and preserves product/fuel, location/time, operating conditions, response and responsible review. Actual digital/peer evidence is retained.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use actual site authorization and exposure programs under qualified supervision. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess correct record/jurisdiction selection and explanation of dust-control components. | 0 / 1 / 2 | Evidence/help: _____ |
| Emission testing, permit interpretation and occupational exposure assessment require the applicable competent professionals. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Determine asphalt mix design

## Worked example · Keep the percentage basis

Fictional teaching JMF uses 5.2% binder by total mixture mass. A 25.0 t batch is considered for arithmetic only. No production temperature or field design is prescribed.

Binder model mass = 25.0 × .052 = 1.30 t; aggregate/other stated remainder = 23.70 t if no other components are separately specified. If the percentage were instead based on aggregate mass, this calculation would be different. Confirm the stated basis before comparing records.

### Explain Marshall and Superpave as different specified design/testing frameworks

Explain Marshall and Superpave as different specified design/testing frameworks. List five JMF fields needed by a production/placement crew.

**Answer guidance.**

Identify specimen/compaction and evaluation differences at a conceptual level; fields include formula/version, sources, mixture type, binder grade/content/basis, recovered inputs and governing requirements.

**Observe:** Reading, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Recalculate binder model mass for 32.0 t at 5.2%

Recalculate binder model mass for 32.0 t at 5.2%. Explain why changing the mass percentage in the field is not an authorized design change.

**Answer guidance.**

32 × .052 = 1.664 t. A change requires the specified technical/contract process; appearance alone is insufficient.

**Observe:** Numeracy, Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

### Build a digital JMF-reading card and have a partner swap the version identifier

Build a digital JMF-reading card and have a partner swap the version identifier. Revise the delivery check and record what must be confirmed.

**Answer guidance.**

Card connects the current approved formula with the load and tests. A different identifier triggers clarification/quality disposition rather than silently reusing old values.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use an authorized real JMF or the complete fictional arithmetic card. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess accurate identification of mixture, design method, percentage basis and temperature measurement requirements. | 0 / 1 / 2 | Evidence/help: _____ |
| Mix design approval and changes are assessed by the qualified design/quality process. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Confirm that samples match mix design

## Worked example · A result with a broken chain

Fictional bag label says aggregate bin A, 09:15. The sampling log says bin B, 09:15, and the test sheet uses the A label. The test values are within the limits printed on the teaching card.

The identity conflict prevents a reliable match to the intended material. Preserve all records, hold the interpretation and resolve the sampling chain through the responsible quality role. Passing numbers do not repair identity. If the source cannot be established, the required resampling/disposition process must be used.

### Build a sample route from planned location to test report

Build a sample route from planned location to test report. Identify where representativeness, safety and identity can each fail.

**Answer guidance.**

Include approved plan/location, safe collection, increments/reduction, container/label, transfer, laboratory receipt and result. Explain distinct failures at relevant points.

**Observe:** Reading, Problem solving, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32

Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32. Explain why a 140°C surface reading is not proof of the internal temperature.

**Answer guidance.**

284°F. The conversion preserves the measured surface quantity; internal temperature requires the appropriate method/location. No field temperature requirement is implied.

**Observe:** Numeracy, Writing through the learner's actual contribution, calculation, explanation or saved work.

### Save a sample register for the worked case

Save a sample register for the worked case. A partner supplies a corrected log with an author/date; retain the original and document the resolution rather than overwriting history.

**Answer guidance.**

Record preserves the discrepancy, authorized correction, evidence and disposition. Digital versioning and individual peer feedback are observed.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use the actual approved sampling and hazardous-energy/access procedures with qualified supervision. | 0 / 1 / 2 | Evidence/help: _____ |
| Observe representative collection/reduction, contamination prevention, labelling and temperature method/recording. | 0 / 1 / 2 | Evidence/help: _____ |
| Do not stage live-belt access or hot-material handling without the complete authorized arrangement. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Determine haul truck samples

## Worked example · Random selection with traceability

Fictional teaching plan identifies loads 1–10 and a recorded random selection of loads 3 and 8. Load 8 is delayed; load 7 is easier to reach. The sampler proposes silently substituting 7.

The approved selection has changed. Follow the plan's authorized contingency/rescheduling method and record any change; do not silently substitute a convenient load. Maintain safe coordination while waiting or arranging the approved alternative. Random selection and safe access must both be satisfied.

### Explain and apply the idea

Prepare the complete label and transfer record for fictional load 3, mix JMF-C revision 2, sampled at 10:25 for binder/gradation testing. Mark any information the case does not supply.

**Answer guidance.**

Include load/mix/version/time/test and fields for actual location, sampler, method, container/seal and transfer/receipt. Unknown data stays open for confirmation, not invented.

**Observe:** Reading, Writing through the learner's actual contribution, calculation, explanation or saved work.

### Two fictional increments are 6.2 kg and 5.8 kg

Two fictional increments are 6.2 kg and 5.8 kg. Find total mass and explain why this sum alone cannot establish a representative or correctly reduced laboratory sample.

**Answer guidance.**

12.0 kg total. Representativeness depends on the sampling plan and collection; laboratory preparation/reduction follows the method.

**Observe:** Numeracy, Problem solving, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Communicate and verify the work

Create a digital sampling log, introduce the delayed-load change with a partner and preserve the authorized decision trail.

**Answer guidance.**

Log retains original selection, reason for change, authority, actual sample identity and result linkage. Each learner contributes and demonstrates saving/reopening the record.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Instructor establishes actual vehicle separation, access, tools, containers and sampling method with the competent testing role. | 0 / 1 / 2 | Evidence/help: _____ |
| Observe representative sampling from authorized locations, identification, safe handling and backup/transfer procedure. | 0 / 1 / 2 | Evidence/help: _____ |
| Record practical evidence from the actual task; paper labels alone do not demonstrate hot-mix sampling skill. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

# Determine compaction levels

## Worked example · The denominator changes the meaning

Fictional in-place density is 2.34 t/m³. Reference A, maximum theoretical density, is 2.50 t/m³. Reference B, a compacted laboratory reference in a different teaching system, is 2.40 t/m³.

Against A: 2.34/2.50 × 100 = 93.6%. Against B: 2.34/2.40 × 100 = 97.5%. Both calculations use the same numerator; they answer different defined questions. Neither establishes acceptance without the applicable method and criterion. Never relabel 97.5% as percent of maximum theoretical density.

### Explain and apply the idea

Compare gauge and core methods using columns for measurement principle, identification, checks, access and authorization.

**Answer guidance.**

Gauge: calibrated indirect measurement with radiation/licensing requirements. Core: physical specimen under approved sampling/lab method. Both need location, traceability, instrument/method checks and safe access.

**Observe:** Reading, Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

### Repeat the worked ratios for an in-place value of 2.30 t/m³

Repeat the worked ratios for an in-place value of 2.30 t/m³. Write each full label.

**Answer guidance.**

92.0% of reference A; approximately 95.83% of reference B. Different bases must remain explicit; no acceptance threshold is supplied.

**Observe:** Numeracy, Communication through the learner's actual contribution, calculation, explanation or saved work.

### Communicate and verify the work

Build a saved density-result sheet and peer-test whether its denominator, units and test location can be found immediately. Correct any ambiguity.

**Answer guidance.**

Sheet includes actual method/instrument or core ID, reference basis, location/time, units, checks, result and disposition. Peer test and digital actions are recorded individually.

**Observe:** Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

## Check your understanding of density evidence

Five questions, four marks each: 20 marks total. Minimum 14/20 (70%), as required by the outline for this knowledge check. This does not authorize gauge operation.

### Question 1 · Explain one difference between core testing and nuclear-gauge testing, and identify the Canadian authorization issue.

A core is a physical sample tested through its approved method; a nuclear gauge uses radiation-based measurement under its procedure. Canadian operation requires the applicable CNSC licence, training and controls.

**Award one mark for each:** Core method identified; Gauge method identified; Methods not treated as interchangeable; CNSC authorization/training identified.

### Question 2 · Calculate percent density for 2.30 Mg/m³ in place and 2.50 Mg/m³ maximum theoretical.

2.30/2.50 × 100 = 92.0%; reference is maximum theoretical density.

**Award one mark for each:** Correct denominator; Correct calculation; 92.0% with percent unit; Reference named.

### Question 3 · Name four traceability fields needed with a density result.

Examples: location/station, lane/lift, lot/sample ID, date/time, method/instrument ID, density units and reference result.

**Award one mark for each:** One correct field; Second distinct field; Third distinct field; Fourth distinct field.

**Question 4 · A draft changes the reference density to improve the percentage. Explain the response.**

Preserve the original result, verify the required method and reference, reject an unsupported substitution and obtain the responsible quality lead's documented disposition.

**Award one mark for each:** Preserves evidence; Checks required reference; Rejects unsupported substitution; Escalates/documents decision.

**Question 5 · Plan a safe learning observation of core or gauge testing.**

Use the approved sampling/testing plan, controlled access and authorized personnel. Record observation separately from operation and keep method/location/result traceable.

**Award one mark for each:** Approved method/plan; Controlled access; Authorized personnel; Observation/operation distinction.

## Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

| What the observer checks | Result | Evidence |
|---|---|---|
| **Critical:** Safe access, surroundings and communication maintained. | Pass / hold | Observation: _____ |
| **Critical:** Required energy, heat, movement and public-access controls maintained for the actual task. | Pass / hold | Observation: _____ |
| Use a licensed/authorized gauge arrangement or approved core-sampling/laboratory arrangement with qualified supervision. | 0 / 1 / 2 | Evidence/help: _____ |
| Assess method recognition, traceability, calculation and correct interpretation; no learner handles a radiation source without required authorization. | 0 / 1 / 2 | Evidence/help: _____ |
| For the outline knowledge quiz, use the questions in this unit with the provider's grading record; practical testing remains a separately observed outcome. | 0 / 1 / 2 | Evidence/help: _____ |

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

# Trace a material and its sample

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional 1,200 g aggregate sample has retained masses of 180 g, 420 g, 360 g and 240 g in the pan. Its log says Mix M, lot 4, but the container label says lot 5. A draft report says “the sample is representative because it is large” and “20% RAP means 20% lower emissions.” The project location is in Metro Vancouver. No acceptance limits or permit number are supplied.

## Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

**Evidence and feedback:** Use the exact case facts; distinguish observation, assumption and missing authority.

## Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

**Evidence and feedback:** The note preserves verified facts, units and unresolved holds without inventing a release.

## Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

**Evidence and feedback:** Use the worked numerical key below; show the relationship and its practical limits.

## Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

**Evidence and feedback:** Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

## Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

**Evidence and feedback:** Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

## Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

**Evidence and feedback:** Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

## Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

**Evidence and feedback:** Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

## Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

**Evidence and feedback:** Use the change-response key below; retain the earlier version and do not alter unrelated facts.

## Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

**Evidence and feedback:** The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

## Change card · Use after the first attempt

The quality lead supplies a corrected source record linking the container to lot 5, with documented authorization and preserved original record.

## Verified case key

Retained total = 1,200 g; cumulative passing after three sieves = 85%, 50%, 20%. Resolve the lot mismatch; sample size does not establish representativeness. RAP percentage does not establish emission reduction. Confirm the applicable Metro regulatory/permit requirements; do not apply the provincial Asphalt Plant Regulation as if it covered Metro.

## Change-response key

Update the traceability record transparently; do not erase the original discrepancy or infer acceptance limits.

## Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.